Moh Maya

Entity Data Model — V1.0

Technical Design Document • Victus

Purpose. This document defines the initial data model for Moh Maya's core V1 entities. It follows the World & Engine Definition and Resources & States Definition. The model is intentionally minimal and extensible; additional data points may be introduced when development demonstrates a genuine need.

1. Design Principles

• **Entities represent things that exist** in the simulated world.

• **Properties describe entities.** Not every piece of information needs to become a graph node.

• **Relationships connect entities** and are especially important to the world model.

• **States and resources remain conceptually separate** from identity and relationships.

• **Specific items are instantiated when needed** rather than predefining an exhaustive inventory.

• **Extensibility is intentional.** V1 is a foundation, not a permanent schema.

2. Common Entity Foundation

All entity types share a minimal conceptual foundation. The eventual storage representation is deliberately not fixed here.

Field	Purpose
id	Unique identifier.
type	Entity type from the V1 vocabulary.
name / label	Human-readable name or label where applicable.
metadata	Optional supporting information that does not yet justify a dedicated field.

3. V1 Entity Vocabulary

The V1 entity vocabulary is: **World, Location, Person, Household, Organization, Business, Infrastructure, Natural Entity, Item, and Event.**

Entity	Purpose
World	Root context and authoritative representation of the simulated world.
Location	A physical place in the world.
Person	An individual human living in the world.
Household	A group of people sharing a living/economic unit.
Organization	A structured group or institution operating in the world.
Business	An organization participating in economic activity by providing goods or services.
Infrastructure	Physical systems or facilities supporting the world.
Natural Entity	Relevant natural or environmental entities.
Item	A specific physical good or possession instantiated when needed.

Entity	Purpose
Event	A happening with time, participants, causes, state, and consequences.

4. World

Field / Group	Purpose
id	Unique world identifier.
name	World name or identifier.
current_time	Current simulated date/time.
calendar	Calendar context used by the simulation.
world_state	Authoritative aggregate state of the world.
history	Important historical facts and consequences.

5. Location

Field / Group	Purpose
id	Unique location identifier.
name	Location name.
type	Kind of location, such as city, neighborhood, house, café, hospital, road, or park.
parent_location	Containing or geographically associated location where applicable.
state	Relevant current condition.
capacity	Relevant capacity or limits where applicable.
properties	Other location-specific characteristics required by the simulation.

6. Person

Field / Group	Purpose
id	Unique person identifier.
name	Person's name.
age	Current age.
current_location	Current physical location.
attributes	Personality, skills, preferences, and other attributes.
states	Current health, hunger, fatigue, stress, mood, and other relevant states.
resources	Resources available to the person.
roles	Profession, employment, social or institutional roles.
relationships	Connections to other entities.
knowledge	What the person knows, believes, or has learned.
memory	Relevant remembered experiences and established personal history.
history	Established life history and canon facts.
goals / circumstances	Current motivations, obligations, goals, and circumstances relevant to behaviour.

7. Household

Field / Group	Purpose
id	Unique household identifier.
name / label	Optional household identifier.
members	People belonging to the household.
location	Primary residence.
resources	Household-level resources.
states	Relevant household conditions.
obligations	Shared financial or practical obligations.
history	Important established household history.

8. Organization

Field / Group	Purpose
id	Unique organization identifier.
name	Organization name.
type	General organization or institution type.
locations	Relevant physical locations.
members	People associated with the organization.
resources	Resources controlled by the organization.
states	Current organizational or institutional condition.
roles	Relevant roles and responsibilities.
relationships	Connections to people, businesses, government, and other organizations.

9. Business

Field / Group	Purpose
id	Unique business identifier.
name	Business name.
business_type	Kind of business.
location	Primary physical location.
owners	Relevant owners.
employees	People employed by the business.
resources	Money, goods, inventory, equipment, and other relevant resources.
states	Current operational and economic condition.
goods / services	What the business provides.
financial_context	Relevant revenue, costs, prices, and obligations.

10. Infrastructure

Field / Group	Purpose
id	Unique infrastructure identifier.
name	Infrastructure name.
type	Kind of infrastructure.
location / coverage	Where it exists or what area it serves.
capacity	Relevant capacity or limit.
state	Current physical or operational condition.
resources	Relevant resources associated with or controlled by it.
dependencies	Other entities or systems it depends on.

11. Natural Entity

Field / Group	Purpose
id	Unique identifier.
name	Name where applicable.
type	Kind of natural or environmental entity.
location / area	Where it exists or what area it affects.
state	Current environmental or physical condition.
capacity / quantity	Relevant quantity or limit where applicable.
relationships	Connections to locations, infrastructure, resources, and events.

12. Item

Field / Group	Purpose
id	Unique item identifier.
type	Kind of item.
name / description	Human-readable item description.
owner	Current owner where applicable.
location	Current physical location.
condition	Current physical condition.
state	Other relevant item state.
value / price	Relevant economic value where applicable.
properties	Item-specific characteristics required by the simulation.

13. Event

Field / Group	Purpose
id	Unique event identifier.
type	Emergent, scheduled, or God event, plus a more specific type where needed.
time	When the event occurs or its relevant time span.
location	Where it occurs, where applicable.

Field / Group	Purpose
participants	Entities involved.
cause	Known cause or causal conditions.
state	Current event state, such as pending, active, resolved, or cancelled.
consequences	Changes produced by the event.
history / record	Relevant record needed for continuity and causal history.

14. Example Entity Relationships

The model is intended to support a graph-oriented representation. These are examples rather than a complete V1 relationship vocabulary.

Relationship	Example
CONTAINS	World → Location; Location → Business
LOCATED_AT	Person → Location; Item → Location
LIVES_IN	Person → Household
MEMBER_OF	Person → Household / Organization
WORKS_AT	Person → Business / Organization
OWNS	Person / Household / Business → Item
RELATED_TO	Person → Person
KNOWS	Person → Person
PARTICIPATES_IN	Person / Organization → Event
LOCATED_IN	Location / Infrastructure / Natural Entity → broader Location
DEPENDS_ON	Infrastructure / Organization → another entity
SERVES	Infrastructure / Organization → Location / population / organization

15. What Does Not Automatically Become an Entity

Resources, states, attributes, professions, personality, relationships, knowledge, and memories remain associated data or connections rather than automatically becoming standalone entity nodes. A future implementation may promote a concept to a node if the simulation demonstrates a real advantage.

16. Evolution Rule

Add a data point when it is required to represent a meaningful simulation rule, preserve continuity, support an interaction, or maintain authoritative world state. Avoid adding fields merely because they could theoretically be useful.

Guiding principle: Keep the model small enough to understand, but rich enough for the world to behave.

Status: V1.0 — Initial entity data model established. Future additions should be driven by implementation needs.

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